



RTL LABORATORY FILE

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SUBMITTED TO:

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(RTL) WEEK # 04 LAB

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RTL LAB WEEK # 04

DESIGN AND SIMULATION OF VGA CONTROLLER USING SRAM USING SYSTEMVERILOG

➤ OBJECTIVE:

To design and simulate a VGA Controller that reads pixel data from SRAM and generates VGA display signals using SystemVerilog.

LAB TASKS

- **Question 1:** In SystemVerilog design, synthesize, and simulate a Video Graphics Controller (VGA Controller) using SRAM.

❖ CODE:

• DESIGN CODE:

```
// ABDUL RAFAY
// PSEB USTP INSPIRE SESSION
// SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
// WEEK # 04 (RTL)
// CODE TASK # 01
// VIDEO GRAPHICS CONTROLLER (VGA CONTROLLER) USING SRAM
module vga_controller (
    input logic    clk,
    input logic    rst_n,
    input logic [7:0] sram_data,
    output logic [16:0] sram_addr,
    output logic    sram_oe,
    output logic [3:0] vga_r,
    output logic [3:0] vga_g,
    output logic [3:0] vga_b,
    output logic    vga_hsync,
    output logic    vga_vsync,
    output logic    vga_blank
);
    logic [9:0] h_count;
    logic [9:0] v_count;
    logic    h_visible;
    logic    v_visible;
    logic    video_active;
    localparam H_VISIBLE = 640;
    localparam H_FRONT  = 16;
    localparam H_SYNC    = 96;
    localparam H_BACK    = 48;
    localparam H_TOTAL   = 800;
    localparam V_VISIBLE = 480;
    localparam V_FRONT   = 10;
```

```

localparam V_SYNC = 2;
localparam V_BACK = 33;
localparam V_TOTAL = 525;
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        h_count <= 10'd0;
    end else if (h_count == H_TOTAL - 1) begin
        h_count <= 10'd0;
    end else begin
        h_count <= h_count + 10'd1;
    end
end
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        v_count <= 10'd0;
    end else if (h_count == H_TOTAL - 1) begin
        if (v_count == V_TOTAL - 1) begin
            v_count <= 10'd0;
        end else begin
            v_count <= v_count + 10'd1;
        end
    end
end
assign h_visible = (h_count < H_VISIBLE) ? 1'b1 : 1'b0;
assign v_visible = (v_count < V_VISIBLE) ? 1'b1 : 1'b0;
assign video_active = h_visible && v_visible;
assign vga_hsync = ((h_count >= (H_VISIBLE + H_FRONT)) &&
    (h_count < (H_VISIBLE + H_FRONT + H_SYNC))) ? 1'b0 : 1'b1;
assign vga_vsync = ((v_count >= (V_VISIBLE + V_FRONT)) &&
    (v_count < (V_VISIBLE + V_FRONT + V_SYNC))) ? 1'b0 : 1'b1;
assign vga_blank = ~video_active;
assign sram_addr = video_active ? {9'd0, v_count[8:1], h_count[9:1]} : 17'd0;
assign sram_oe = video_active;
assign vga_r = video_active ? sram_data[7:4] : 4'd0;
assign vga_g = video_active ? sram_data[3:2] : 4'd0;
assign vga_b = video_active ? sram_data[1:0] : 4'd0;
endmodule

```

• TEST BENCH CODE:

```

// ABDUL RAFAY
// PSEB USTP INSPIRE SESSION
// SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
// WEEK # 04 (RTL)
// CODE TASK # 01
// TEST-BENCH VIDEO GRAPHICS CONTROLLER (VGA CONTROLLER) USING SRAM
module tb_vga_controller;
    logic    clk;
    logic    rst_n;
    logic [7:0] sram_data;
    logic [16:0] sram_addr;
    logic    sram_oe;

```

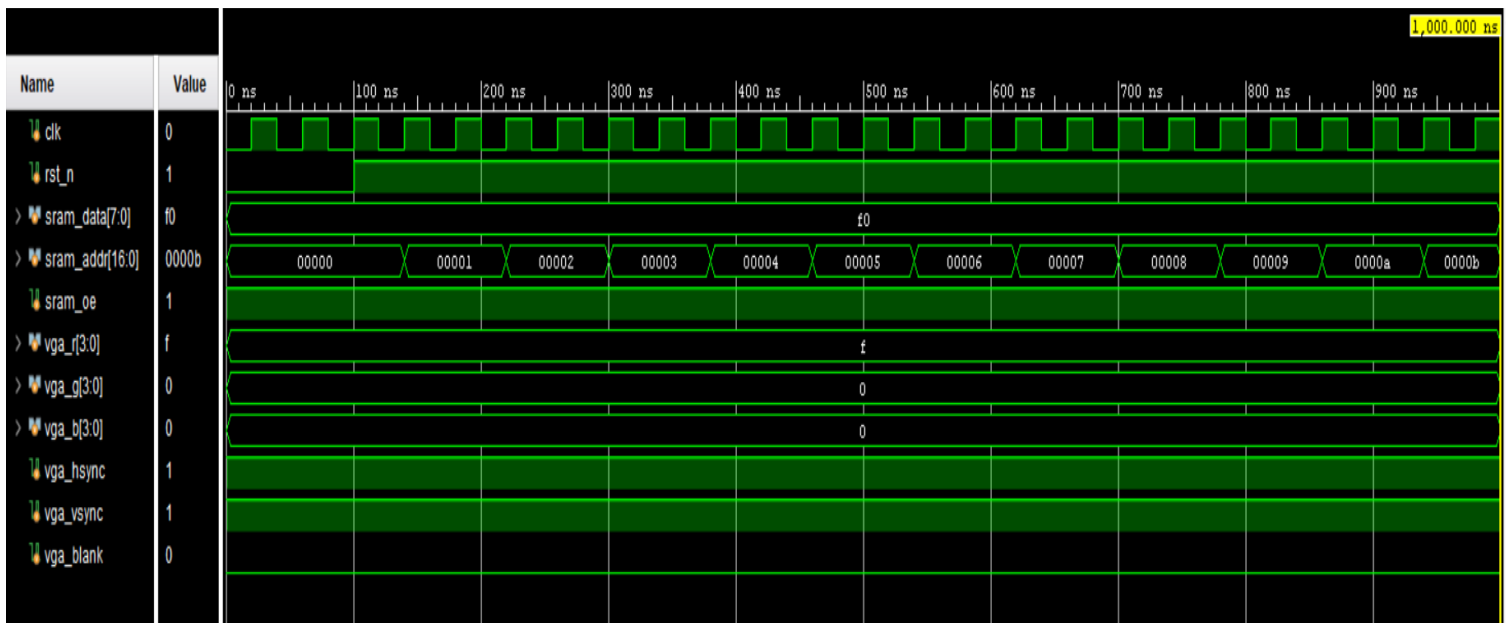
```

logic [3:0] vga_r;
logic [3:0] vga_g;
logic [3:0] vga_b;
logic      vga_hsync;
logic      vga_vsync;
logic      vga_blank;
logic [7:0] sram_mem [0:307199];
vga_controller uut (
    .clk(clk),
    .rst_n(rst_n),
    .sram_data(sram_data),
    .sram_addr(sram_addr),
    .sram_oe(sram_oe),
    .vga_r(vga_r),
    .vga_g(vga_g),
    .vga_b(vga_b),
    .vga_hsync(vga_hsync),
    .vga_vsync(vga_vsync),
    .vga_blank(vga_blank)
);
initial begin
    clk = 0;
    forever #20 clk = ~clk;
end
initial begin
    $dumpfile("vga_controller.vcd");
    $dumpvars(0, tb_vga_controller);
end
initial begin
    integer i;
    for (i = 0; i < 307200; i++) begin
        sram_mem[i] = 8'h00;
    end
    for (i = 0; i < 6400; i++) begin
        sram_mem[i] = 8'hF0;
    end
    for (i = 6400; i < 12800; i++) begin
        sram_mem[i] = 8'h0F;
    end
end
always_comb begin
    if (sram_oe) begin
        sram_data = sram_mem[sram_addr];
    end else begin
        sram_data = 8'h00;
    end
end
initial begin
    rst_n = 1'b0;
    #100 rst_n = 1'b1;
    #2000000 $finish;

```

```
end
endmodule
```

❖ WAVE FORM OUTPUT:



❖ SCHEMATIC OUTPUT:

